

5.5 For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.

$$\text{a. } \text{var}(b_2) = \frac{\sigma^2}{\sum (x_t - \bar{x})^2}, \quad x_t = t.$$

$$\sum (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \frac{\left(\sum_{t=1}^T t\right)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{1}{T} \frac{T^2(T+1)^2}{4} = T(T+1) \frac{2(2T+1) - 3(T+1)}{12} = T(T+1) \frac{T-1}{12} = \frac{T(T^2-1)}{12}$$

$$\text{var}(b_2|\mathbf{x}) = \frac{12\sigma^2}{T(T^2-1)} = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T^2-1)} = 0, \quad b_2 \text{ is a unbiased estimator.}$$

- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1-r^n)}{1-r}$$

$$x_t = 0.5^t. \quad \sum x_t = \sum_{t=1}^T 0.5^t = \frac{0.5(1-0.5^T)}{1-0.5} = 1 - (0.5)^T.$$

$$\sum x_t^2 = \sum_{t=1}^T (0.5^t)^2 = \sum_{t=1}^T 0.5^{2t} = \sum_{t=1}^T (0.25)^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{1-(0.25)^T}{3}$$

$$\sum (x_t - \bar{x})^2 = \sum_{t=1}^T x_t^2 - \frac{\left(\sum_{t=1}^T x_t\right)^2}{T} = \frac{1}{3}(1-(0.25)^T) - \frac{1}{T}(1-(0.5)^T)^2.$$

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{3}(1-(0.25)^T) - \frac{1}{T}(1-(0.5)^T)^2}. \quad \lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{3} - 0} = 3\sigma^2 \neq 0, \quad b_2 \text{ is NOT a unbiased.}$$

- c. Provide an intuitive explanation for these results.

In (a), $x_t = t$ spread out continuously over time, the $\text{var}(b_2|\mathbf{x})$ larger and larger, more samples can offer effective information about β_2 .

In (b), $x_t = 0.5^t$ decays exponentially, early observations carry some information, but later observations all collapse toward 0, contributing virtually no new information.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

- a. $H_0: \beta_2 = 0 \quad H_1: \beta_2 \neq 0. \quad \alpha = 0.05$

$$t^* = \frac{3-0}{\text{se}(b_2)} = \frac{3}{\sqrt{4}} = 1.5. \quad RK = \{t^* | |t^*| > t_{0.025, 60}\} = \{t^* | t^* > 2\}$$

' $|t^*| = 1.5 < 2$ & RK, we don't have enough evidence supports that $\beta_2 \neq 0$. #

- b. $H_0: \beta_1 + 2\beta_2 = 5 \quad H_1: \beta_1 + 2\beta_2 \neq 5. \quad \alpha = 0.05$

$$\text{var}(b_1 + 2b_2) = \text{var}(b_1) + 4\text{var}(b_2) + 4\text{cov}(b_1, b_2) = 3 + 4 \times 4 + 4 \times (-2) = 3 + 16 - 8 = 11$$

$$\text{se}(b_1 + 2b_2) = \sqrt{11}. \quad t^* = \frac{2+6-5}{\sqrt{11}} = \frac{3}{\sqrt{11}} = 0.905, \quad RK = \{t^* | |t^*| > t_{0.025, 60}\} = \{t^* | t^* > 2\}$$

$t^* = 0.905 < 2 \Rightarrow$ fail to reject H_0 . #

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$, $\alpha = 0.05$

若 $\hat{\beta} = b_1 - b_2 + b_3$, $\hat{\beta} = 2 - 3 + (-1) = -2$. $\text{var}(\hat{\beta}) = W' \Sigma W$. 其中 $W' = (1, -1, 1)$

$W' \Sigma = (1, -1, 1) \begin{bmatrix} 3 & 4 & 0 \\ 4 & 9 & 0 \\ 1 & 0 & 3 \end{bmatrix} = (7+2+1, -2-4+0, 1+0+3) = (10, -6, 4)$

$W' \Sigma W = (10, -6, 4) \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 10 + 6 + 4 = 16 = \text{var}(\hat{\beta})$

$t^* = \frac{\hat{\beta} - \lambda}{\text{se}(\hat{\beta})} = \frac{-2-4}{\sqrt{16}} = \frac{-6}{4} = -1.5$, $RR = \{t^* | |t^*| > t_{0.025, 60}\} = \{t^* | t^* > 2\}$

$|t^*| = 1.5 < 2 \notin RR$. fail to reject H_0 #

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2, \text{mean}} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

分前半、後半一半

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1-MR6 hold:

a. Show that $\hat{\beta}_{2, \text{mean}}$ is unbiased.

step 1. 展开 $\bar{y}_1, \bar{y}_2$

$$\bar{y}_1 = \frac{\sum y_i}{N} = \frac{1}{N} \sum (\beta_1 + \beta_2 x_i + e_i) = \frac{1}{N} (N\beta_1 + \beta_2 \sum x_i + \sum e_i) = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$$

$$\bar{y}_2 = \frac{\sum y_i}{N} = \frac{1}{N} \sum (\beta_1 + \beta_2 x_i + e_i) = \frac{1}{N} (N\beta_1 + \beta_2 \sum x_i + \sum e_i) = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

step 2. 将 $\bar{y}_1, \bar{y}_2$ 代入

$$\hat{\beta}_{2, \text{mean}} = \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \frac{\beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

step 3. 取期望值

$$E\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = \beta_2 + \frac{E(\bar{e}_2) - E(\bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \quad (\because x \text{ is given, } \beta_2 \text{ is parameter})$$

又根据 MR2: $E(e_i | x) = 0$, 故 $E\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = \beta_2$, $\hat{\beta}_{2, \text{mean}}$ is unbiased. #

b. Derive an expression for $\text{var}(\hat{\beta}_{2, \text{mean}} | x)$.

$$\text{var}(\hat{\beta}_2 | x) = \text{var}\left(\frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}\right) = \left(\frac{1}{\bar{x}_2 - \bar{x}_1}\right)^2 \text{var}(\bar{y}_2 - \bar{y}_1) = \left(\frac{1}{\bar{x}_2 - \bar{x}_1}\right)^2 \text{var}(\bar{e}_2 - \bar{e}_1)$$

$\hookrightarrow \text{var}(\bar{e}_2) + \text{var}(\bar{e}_1) - 2\text{cov}(\bar{e}_1, \bar{e}_2)$

又根据 MR4: $E(e_i, e_j | x) = 0$, $i \neq j$. 及 MR3: $\text{var}(e_i | x) = \sigma^2$.

且 $\bar{e}_1$ 样本数 $= \frac{N}{2} = \bar{e}_2$ 样本数. 若 $\frac{N}{2} = m$, 则 $\text{var}(\bar{e}_1) = \text{var}\left(\frac{1}{m} \sum e_i\right) = \frac{1}{m^2} \sum \text{var}(e_i) = \frac{1}{m^2} (m\sigma^2) = \frac{\sigma^2}{m}$,

同理 $\text{var}(\bar{e}_2) = \frac{\sigma^2}{m}$,

故 $\text{var}(\hat{\beta}_2 | x) = \left(\frac{1}{\bar{x}_2 - \bar{x}_1}\right)^2 [\text{var}(\bar{e}_2) + \text{var}(\bar{e}_1)] = \left(\frac{1}{\bar{x}_2 - \bar{x}_1}\right)^2 \left(\frac{2\sigma^2}{m}\right) = \frac{2\sigma^2}{m(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$ #

c. Write down an expression for $\text{var}(\hat{\beta}_{2, \text{mean}})$.

$$\text{var}(\hat{\beta}) = E_x[\text{var}(\hat{\beta} | x)] + \text{var}_x[E(\hat{\beta} | x)]$$

在 x is given 下, $\hat{\beta}_{2, \text{mean}}$ 是 β_2 的不偏估计量, 故 $E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2 \Rightarrow \text{var}_x[E(\hat{\beta}_2 | x)] = \text{var}_x(\beta_2) = 0$ ($\because \beta_2$ 为常数).

故 $\text{var}(\hat{\beta}_{2, \text{mean}}) = E_x\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right)$ #

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?

由(1)知, $Var(\hat{\beta}_{2,mean}) = E_x\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right)$, 且 $\hat{\beta}_{2,mean}$ is unbiased.

故当 $N \rightarrow \infty$, $Var(\hat{\beta}_{2,mean}) \rightarrow 0$

条件: $\bar{x}_2 - \bar{x}_1 \neq 0 \Rightarrow$ MRS: 解释变数间不能有完全线性关系.

只要解释变数是從一個具有常數且大於零的變異數的母體中抽樣出來的, 那 $\hat{\beta}_{2,mean}$ 就會是 β_2 的估計量

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

- i. $var(b_2|x)$ and $var(\hat{\beta}_{2,mean}|x)$ where b_2 is the OLS estimator.
- ii. Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

當樣本數 N = 100 時的計算結果： =====	當樣本數 N = 1000 時的計算結果： =====
[Part i] 給定單一組 x 的條件變異數： var(b2 x) = 1.116916 var(b2_mean x) = 1.48638 -> 驗證：LSE 的變異數確實比較小 (高斯-馬可夫定理)	[Part i] 給定單一組 x 的條件變異數： var(b2 x) = 0.1237935 var(b2_mean x) = 0.1652595 -> 驗證：LSE 的變異數確實比較小 (高斯-馬可夫定理)
[Part ii] 進行 1000 次模擬所估計出來的期望值 (無條件變異數的核心)： E[(s_x^2)^-1] = 0.1223983 E[4 / (x2_bar - x1_bar)^2] = 0.1653223 -> 觀察：當 N 越來越大時，這兩個期望值會收斂在固定的常數附近。	[Part ii] 進行 1000 次模擬所估計出來的期望值 (無條件變異數的核心)： E[(s_x^2)^-1] = 0.1199984 E[4 / (x2_bar - x1_bar)^2] = 0.1601359 -> 觀察：當 N 越來越大時，這兩個期望值會收斂在固定的常數附近。
當樣本數 N = 500 時的計算結果： =====	當樣本數 N = 5000 時的計算結果： =====
[Part i] 給定單一組 x 的條件變異數： var(b2 x) = 0.2397738 var(b2_mean x) = 0.3176633 -> 驗證：LSE 的變異數確實比較小 (高斯-馬可夫定理)	[Part i] 給定單一組 x 的條件變異數： var(b2 x) = 0.02421627 var(b2_mean x) = 0.03234208 -> 驗證：LSE 的變異數確實比較小 (高斯-馬可夫定理)
[Part ii] 進行 1000 次模擬所估計出來的期望值 (無條件變異數的核心)： E[(s_x^2)^-1] = 0.1203267 E[4 / (x2_bar - x1_bar)^2] = 0.1608477 -> 觀察：當 N 越來越大時，這兩個期望值會收斂在固定的常數附近。	[Part ii] 進行 1000 次模擬所估計出來的期望值 (無條件變異數的核心)： E[(s_x^2)^-1] = 0.1199576 E[4 / (x2_bar - x1_bar)^2] = 0.1599683 -> 觀察：當 N 越來越大時，這兩個期望值會收斂在固定的常數附近。

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

Relative magnitudes: For all sample sizes ($N=100$ to 5000), $Var(b_2|x)$ is consistently smaller than $Var(\hat{\beta}_{2,mean}|x)$ (e.g., $1.11 < 1.48$ at $N=100$). This empirically proves the Gauss-Markov Theorem, confirming that LSE is the best linear unbiased estimator (BLUE) with the minimum variance.

Change as N increases: As N increases from 100 to 5000, both conditional variances decrease significantly (e.g., $Var(\hat{\beta}_{2,mean}|x)$ drops 1.48 to 0.032).

Consistency: The data confirms that $\hat{\beta}_{2,mean}$ is a consistent estimator, as its variance converges to zero when N grows larger.

g. Show that $E[(s_x^2)^{-1}] \xrightarrow{P} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{P} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]

$Var(x) = \frac{(b-a)^2}{12}$, $(a,b) = (0,10) \Rightarrow Var(x) = \frac{10^2}{12} = \frac{100}{12}$

隨 $N \rightarrow \infty$, s_x^2 會收斂到母體 variance, 故 $s_x^2 \xrightarrow{P} \frac{100}{12}$

$(s_x^2)^{-1} \xrightarrow{P} \frac{12}{100} = 0.12$ #

当 $N \rightarrow \infty$, 前半段資料會均勻落在 $(0,5)$ 之間; 後半段資料會均勻落在 $(5,10)$ 之間

而前半段平均數 $\bar{x}_1 \xrightarrow{P} E[x|0 < x < 5] = \frac{0+5}{2} = 2.5$, 後半段平均數 $\bar{x}_2 \xrightarrow{P} E[x|5 < x < 10] = \frac{5+10}{2} = 7.5$

$\bar{x}_2 - \bar{x}_1 \xrightarrow{P} 7.5 - 2.5 = 5$ 代回 $\Rightarrow \frac{4}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4}{25} = 0.16$ #

- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

若无按照大小排序，那前半 $\bar{x}_1 \rightarrow 5$ ，後半 $\bar{x}_2 \rightarrow 5$ ， $(\bar{x}_2 - \bar{x}_1) \rightarrow 0$

$$\text{var}(\hat{\beta}_{2,mean}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad , \quad \text{若 } (\bar{x}_2 - \bar{x}_1) \rightarrow 0, \text{ 那 } \text{var.} \rightarrow \infty \quad \#$$